



Keynes Consulting



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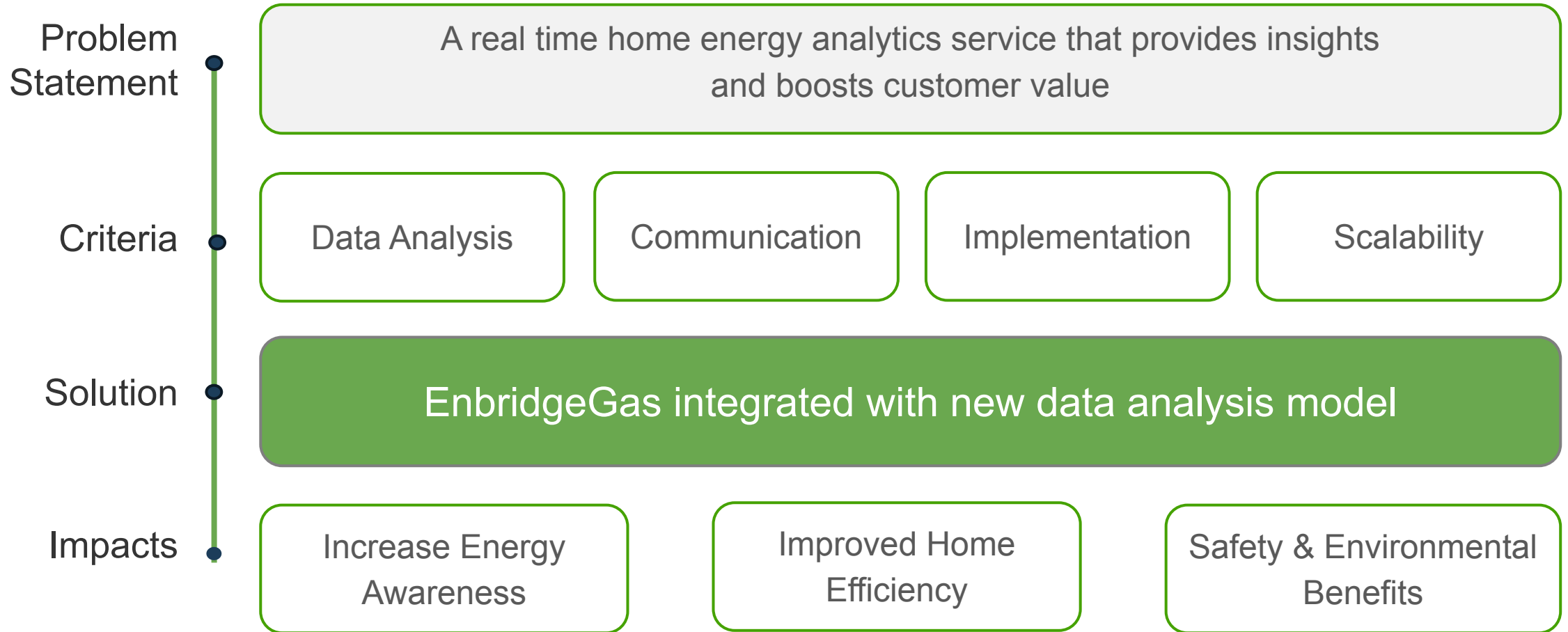


Christian Arteaga

How can Enbridge implement a
data-driven system that empowers
homeowners to better understand
their gas **consumption**?

Executive Summary

Agenda overview



Analysis

Solution

Implementation

Impact

Homeowner Persona

Identifying Household Challenges

John



Age: 72 (Retired Homeowner)

- Loyal Enbridge customer for 40 years
- Concerned about rising utility costs
- Skeptical about the new technology

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Homeowner Persona

Identifying Household Challenges

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The Smith Family



Recently bought a new house:

- Looking for a reliable energy provider
- Cautious of their carbon footprint
- Prioritizing efficient energy usage

Analysis

Solution

Implementation

Impact

People, Process, Technology

Current State

Homeowners cannot easily understand or make use of their energy data

Lack of awareness on energy saving

Desired State

Analysis

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People, Process, Technology

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People, Process, Technology

Current State

Homeowners cannot easily understand or make use of their energy data

Lack of awareness on energy saving

Automation + Interactive Interface

Desired State

Better energy usage understanding

Proactive action to ensure gas saving

Analysis

Solution

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People, Process, Technology

Key Takeaway

Current State

Unavailability of energy usage
data

Lack of awareness on energy
saving

Automation + Interactive Interface

People are unable to analyze or
use their energy usage data

Desired State

Better energy usage
understanding

Proactive action to ensure gas
saving

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People, **Process**, Technology

Current State

Manual meter checks

Lack of feedback to clients

Desired State

Analysis

Solution

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People, **Process**, Technology

Current State

Manual meter checks

Lack of feedback to clients

Desired State

Automated and accurate
usage checks

Detailed analysis of usage available
to end users

Analysis

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People, **Process**, Technology

Current State

Manual meter checks

Lack of feedback to clients

Educating customers on new Enbridge data storage and analysis

Desired State

Automated and accurate
usage checks

Detailed analysis of usage available
to end users

Analysis

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People, Process, Technology

Key Takeaway

Implement automated data collection, processing to make it ready to use for clients and users

Current State

Manual monthly checks

Lack of feedback to clients

Educating customers on new Enbridge data storage and analysis

Desired State

Automated and accurate
usage checks

Detailed analysis of usage available
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Analysis

Solution

Implementation

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People, Process, Technology

Current State

Outdated meters

Outdated app with limited features

Desired State

Analysis

Solution

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People, Process, Technology

Current State

Outdated meters

Outdated app with limited features

Desired State

Smart meters

Stronger software infrastructure

Analysis

Solution

Implementation

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People, Process, Technology

Current State

Outdated meters

Outdated app with limited features

Build a healthier software infrastructure

Desired State

Smart meters

Stronger software infrastructure

Analysis

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People, Process, Technology

Key Takeaway

Current State

Outdated meters

Outdated app with limited features

Enbridge software and technology
can be improved for effectiveness

Desired State

Smart meters

Stronger software infrastructure

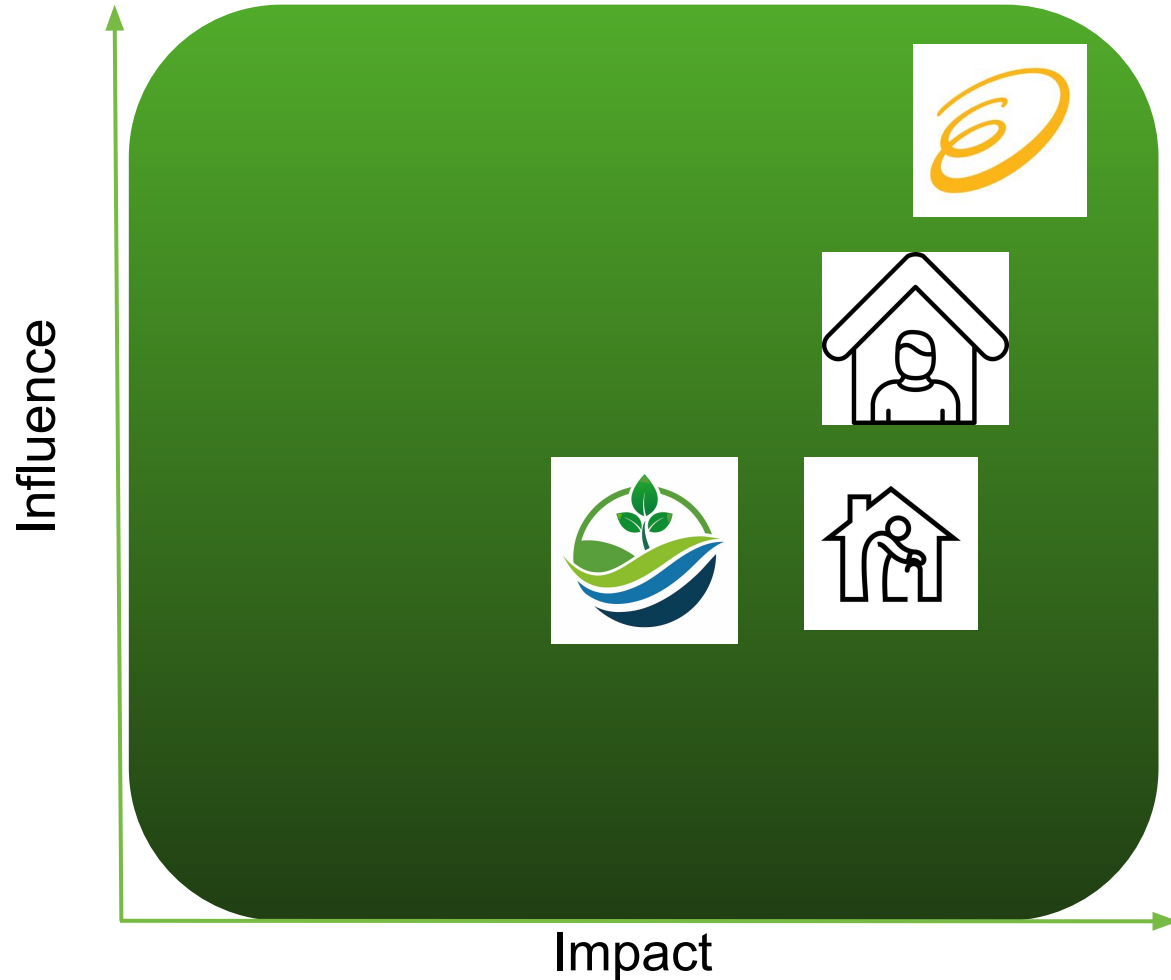
Analysis

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Stakeholder Analysis



Old Homeowners



New Homeowners



Enbridge



Environment

Analysis

Solution

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Stakeholder Analysis

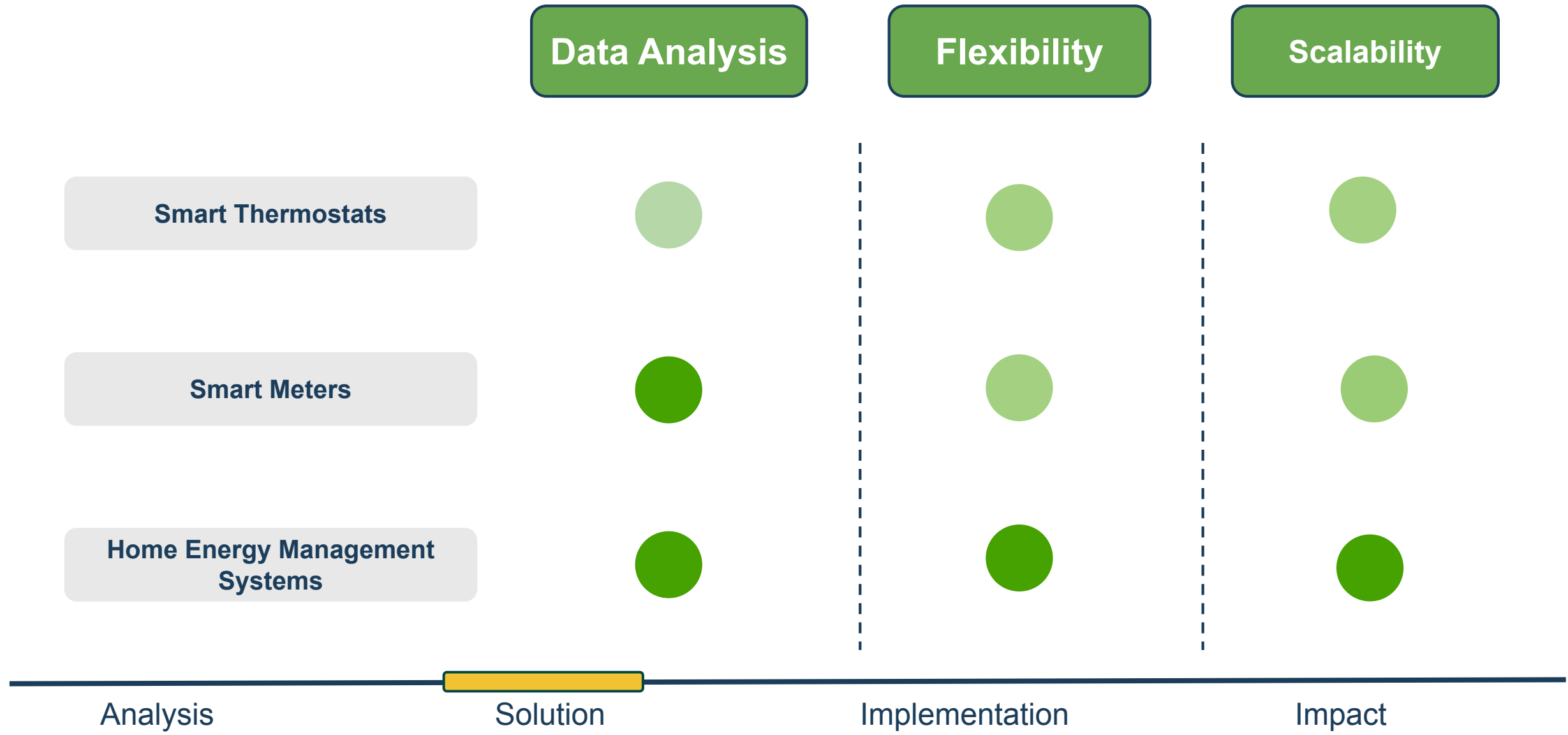
Key Takeaway

The solution must be financially feasible to ensure all parties are not negatively affected



Alternatives

Assessing potential strategies against criteria



Alternatives

Assessing potential strategies against criteria

Data Analysis

Flexibility

Scalability

Smart Thermostats



Smart Meters



**Home Energy Management
Systems**



Analysis

Solution

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1

Initialization Phase

- Redesign Enbridge app
- Develop and Train predictive models
- Implement secure data collection and storage

2

Integration Phase

- Implement HEMS in homes
- Promote the software platform
- Leverage collected data for optimization

Data Collection and Storage

How to collect and store the data

Data Collection



- Real-time energy monitoring
- Pattern detection
- Integration with HEMS

Data Storage



- Cloud-based storage (AWS S3)
- Centralized historical records
- Secure access and backups

Analysis

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Data Analysis

How we can analyze our data

Machine Learning Predictive Modeling

- Generate monthly bill predictions
- Detect anomalies

Time of Use & Behaviour Analysis

- Identifies when homeowners use the most energy
- Detects behaviour patterns: morning spikes, night usage, weekend differences
- Links usage to occupancy patterns

Weather vs Occupancy Adjusted Load Visualization

- Track occupancy patterns
- Compare consumption to outdoor temperature
- Detect wasteful patterns

Behaviour Recommendations

- Create if-then rules from consumption patterns
- Build “Daily Pattern profiles”
- Provide personalized recommendations

Analysis

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Communication to Homeowners

Advertising our new software



- Simple and clean interface
- Low learning curve
- Visually clear energy insights
- Appeals to a wide audience
- Strong UI/UX allows us to clearly communicate our software's value

Analysis

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New System Implementation - ADKAR

How we advertise our new software

Awareness

“The current system works fine, why do we need to change it?”

Growing security threats and aging equipment make this change necessary.

Analysis

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New System Implementation - ADKAR

How we advertise our new software

Awareness

“The current system works fine, why do we need to change it?”

Growing security threats and aging equipment make this change necessary.

Desire

“Why should I have to learn how to use this new technology?”

Lower your monthly bills and gives you more control over your home's energy use

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Knowledge

“How hard is it to understand my own data?”

Clear visuals and simple insights make your energy data easy to understand

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Ability

“What if I'm not good with technology?”

Step-by-step guides and intuitive design make the system easy for anyone to use

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Ability

“What if I’m not good with technology?”

Step-by-step guides and intuitive design make the system easy for anyone to use

Reinforcement

“What if I forget how to use it later?”

Ongoing tips and built-in support ensure homeowners stay comfortable and confident

Analysis

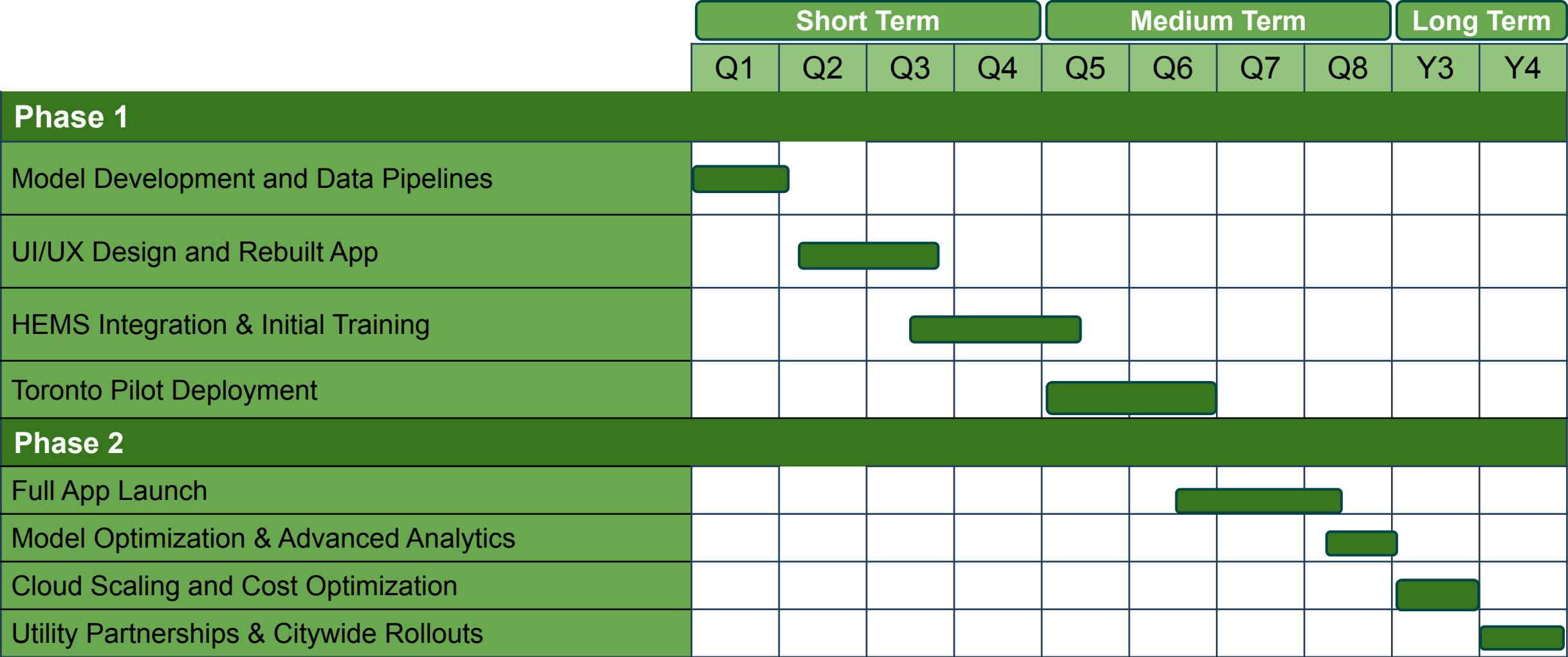
Solution

Implementation

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Timeline

Key tasks associated with the success of the implementation



Financials

Where our money will go

HEMS hardware installation	~600 million (~500 per home)
Software Licensing	~35 million (~30 per home)
Cloud Storage & Analytics	~12 million (~10 per home)
Marketing and Onboarding (one time)	~35 million (~30 per home)
Maintenance & support per year	~35 million (~30 per home)
Model Development	~1 million

Initial Costs: 718 million

Ongoing Annual Costs: 82 million

Analysis

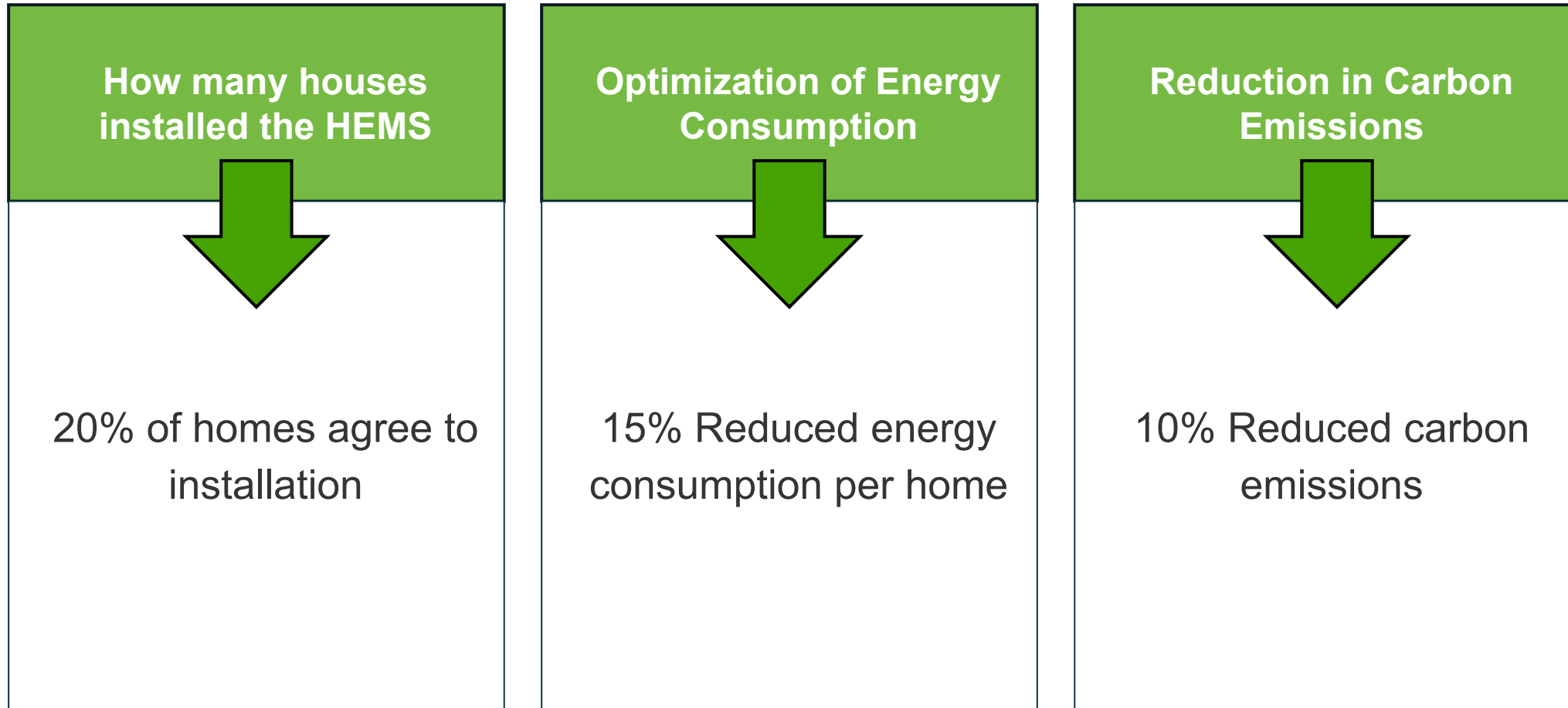
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Key Performance Indicators

Assessing the performance of the implementation



Analysis

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Implementation

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Personal Impacts

Back to John and the Smith Family

John



Age: 72 (Retired Homeowner)

- | | | |
|--|---|--|
| ● Loyal Enbridge customer for 40 years | → | ● No more manual meter checks |
| ● Concerned about rising utility costs | → | ● Finally cuts down his utility costs with the new tech |
| ● Skeptical about the new technology | → | ● Simple design and training made him feel confident using the app |

Analysis

Solution

Implementation

Impact

Personal Impacts

Back to John and the Smith Family

The Smith Family



Recently bought a new house

- | | | |
|--|---|--|
| ● Looking for a reliable energy provider | → | ● The trust the new tech because Enbridge has been around for more than 75 years |
| ● Cautious of their carbon footprint | → | ● Gives them a clear snapshot of their carbon footprint |
| ● Prioritizing efficient energy usage | → | ● App helps them tweak their habits before the bill arrives |

Analysis

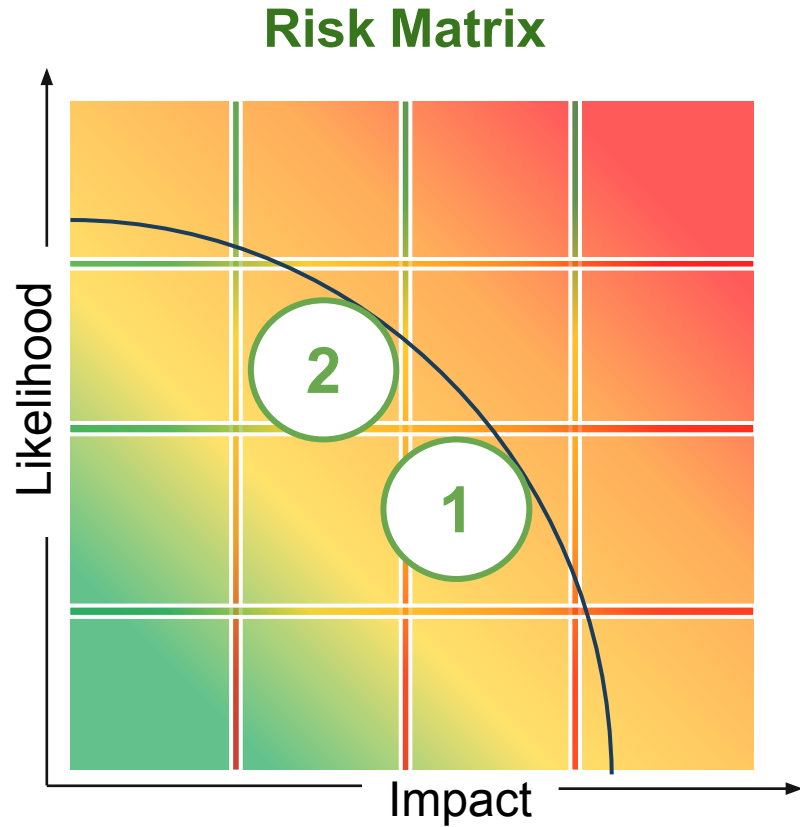
Solution

Implementation

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Risks and Mitigations

Proactive Control



Risk

1 Collecting household energy data

2 High Upfront Integration Costs

Mitigation

- Use end to end encryption
- Provide homeowners with privacy dashboards

- Start with Pilot Program
- Apply for funding from government

Analysis

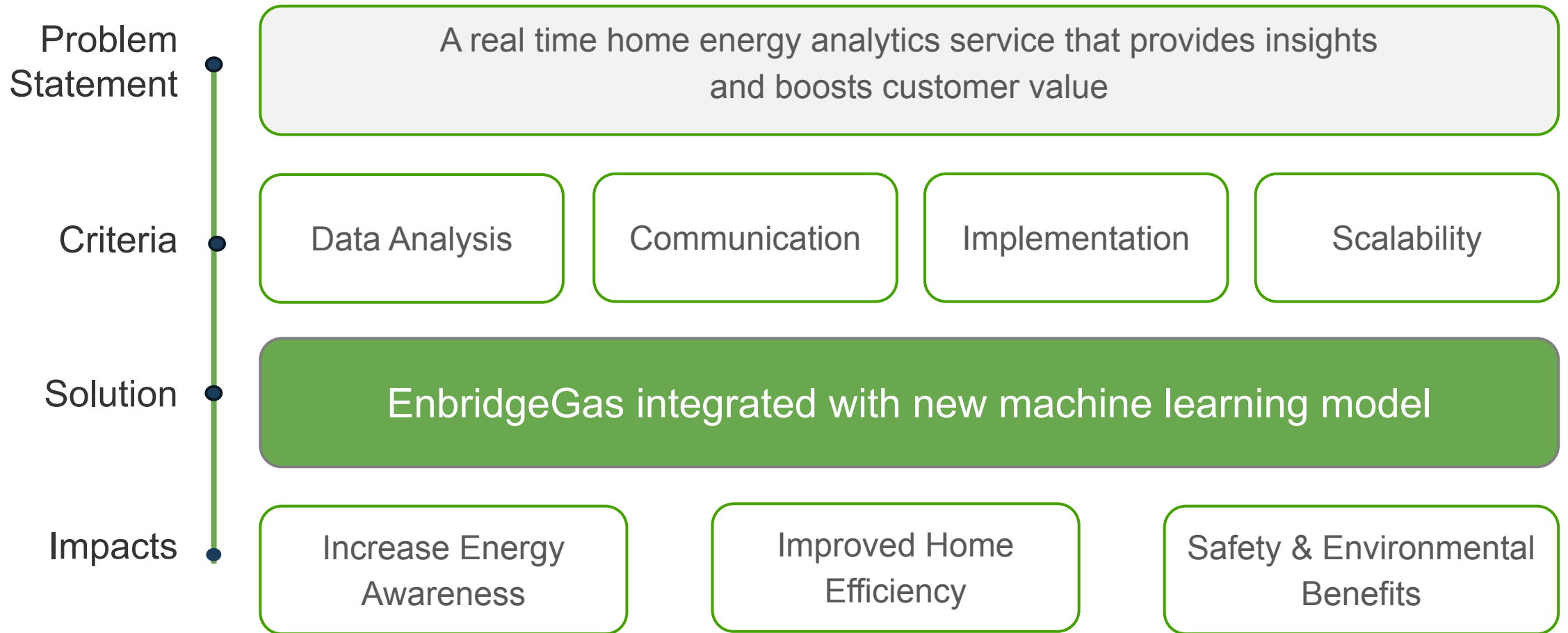
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Q&A

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